

Causal Inference and ML: Beyond Correlation in Sports Data

Estimating Heterogeneous Effects of VO2 Max on Performance

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Github Repository:

`github.com/kboroz/Causal_Inference_and_ML/Week07_Project_Session`

Analytical Framework: VO2 Max Causal Impact

Data & Variables

Treatment (T):

- VO2 Max: Interval

Outcome (Y):

- Medal Won: Binary

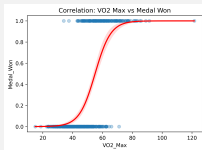
Features (X):

- Experience: Categorical
- Age: Continuous
- Altitude: Binary

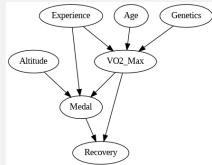
Collider (Z):

- Recovery: Influenced by both T and Y

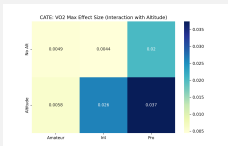
Regression Baseline



Structural Model



CATE: Alti. and Exp.



Key Findings

Every unit increase in VO2-Max raises win probability by $\sim 2.3\%$.